Thermodynamic description of the Al–Zn system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Al–Zn binary system as calculated with pycalphad from the SGTE dataset AlZn-93Mey-LB.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L → (Al) + (Zn), lies at 380.8 °C.

1. Introduction

The Al–Zn system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as simple eutectic, solid miscibility gap, monotectoid. The thermodynamic description is the SGTE collection entry [1], which reproduces the assessment of S. an Mey, Z. Metallkd., 84, 451-55(1993). [2]. The calculations use pycalphad 0.11.2 at 1 atm.

For Al, the liquidus meets the axis at 660.3 °C. For Zn, the liquidus meets the axis at 419.5 °C.

2. Phase diagram and invariant reactions

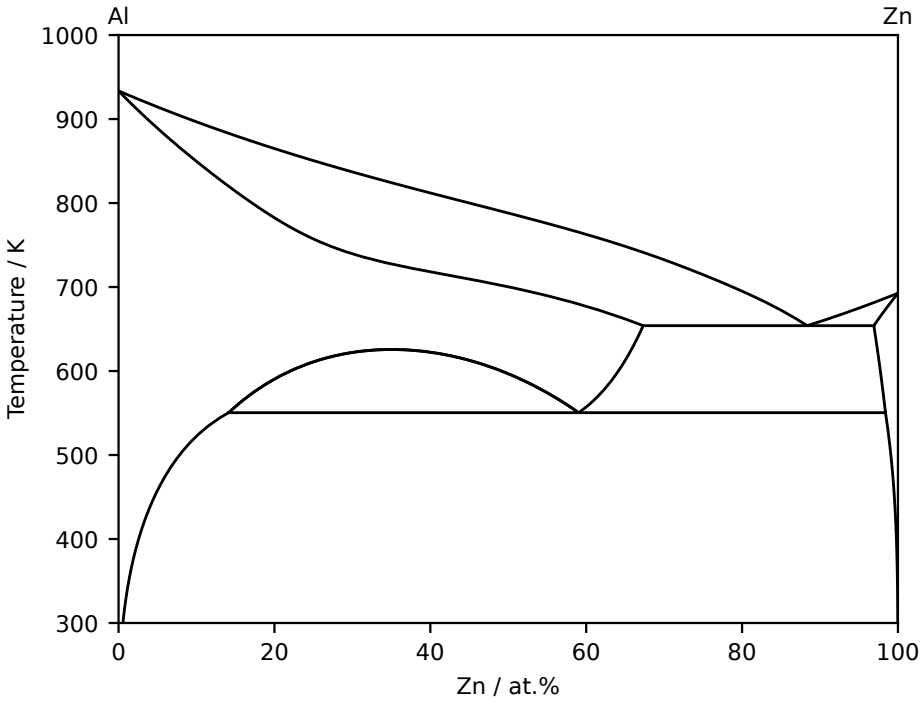


Fig. 1. Calculated Al–Zn phase diagram.

The eutectic reaction L → (Al) + (Zn) is calculated at 380.8 °C, with L at x(Zn) = 0.884. The dataset places a eutectoid ((Al)2 → (Al)1 + (Zn)) at 277.2 °C; the composition of (Al)2 is a mole fraction of 0.590.

Table 1. Invariant reactions and congruent points, calculated (Al–Zn).

Reaction	Type	T / °C	Compositions (at.% Zn)		
		this work	phase 1	phase 2	phase 3
L → (Al) + (Zn)	eutectic	380.8	88.4	67.3	96.9
(Al)2 → (Al)1 + (Zn)	eutectoid	277.2	59.0	14.1	98.4

Phases 1–3 in the order written in the reaction. Calculated with pycalphad 0.11.2 from the SGTE dataset; synthetic document.

3. Phase stability

Table 2. Solid phases: structure, stability range and homogeneity range (Al–Zn).

	Structure			Stability range / K		Homogeneity range	
Phase	Pearson	Strukturb.	Prototype	from	to	x(Zn)	at T / K
(Al)	cF4	A1	Cu	< 300[a]	933.5	0.000–0.671	650
(Zn)	hP2	A3	Mg	< 300[a]	692.7	0.970–1.000	650

[a] Stable at the lowest calculated temperature.

The upper stability limit of (Al) is 660.3 °C. The solubility of Al in (Zn) reaches a mole fraction of 0.031 at 380.8 °C.

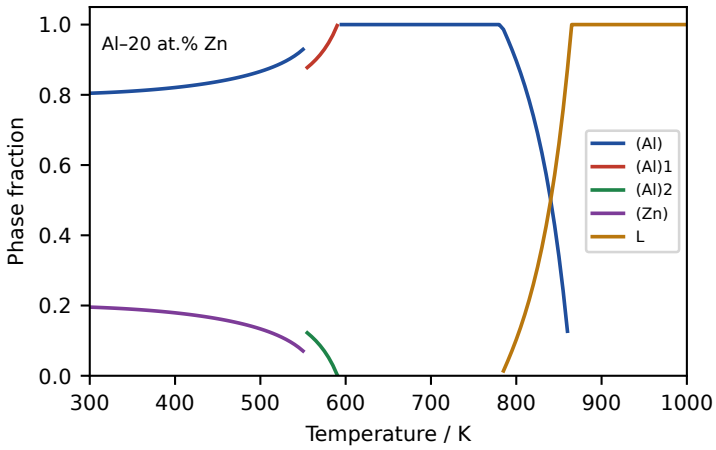


Fig. 2. Calculated equilibrium phase fractions versus temperature for Al–20 at.% Zn.

4. Thermodynamic properties

At 1000 K the integral mixing enthalpy of the liquid has its maximum, 2616 J/mol, at x(Zn) = 0.500. For the equiatomic liquid at 1000 K, the Gibbs energy of mixing is −3.99 kJ/mol.

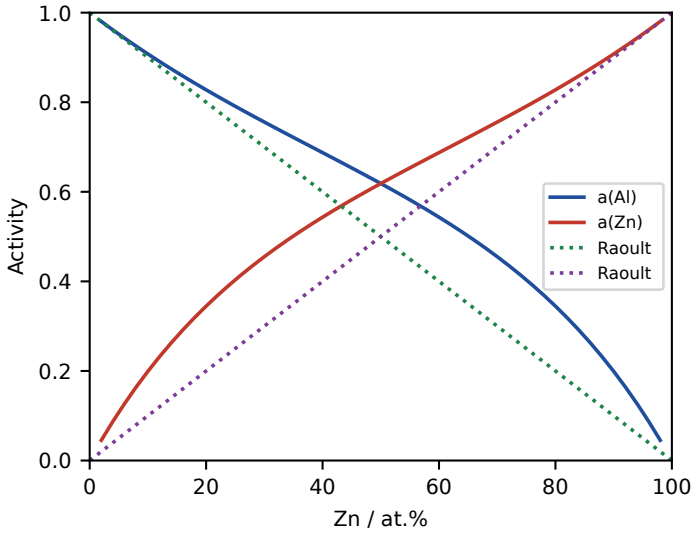


Fig. 3. Calculated activities in the liquid at 1000 K (reference: pure liquids).

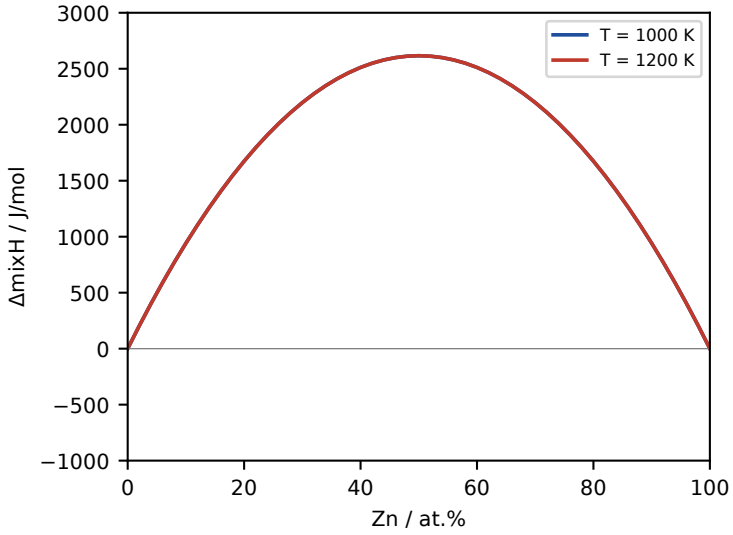


Fig. 4. Calculated mixing enthalpy of the liquid at 1000 K and 1200 K.

Table 3. Redlich–Kister parameters of the liquid (Al–Zn).

	Liquid: L(ν) = a + b·T + c·T·ln T		
ν	**a**	**b**	**c**
L0	10465.55	-3.39259	0.0

Values as printed in the TDB file.

Table 4. Integral and partial mixing properties of the liquid at 1000 K (Al–Zn).

			Liquid at 1000 K			
x(Zn)	ΔmixH / kJ mol⁻¹	ΔmixG / kJ mol⁻¹	a(Al)	a(Zn)	ΔH(Al) / kJ mol⁻¹	ΔH(Zn) / kJ mol⁻¹
0.100	0.94	-2.07	0.9077	0.1992	0.10	8.48
0.200	1.67	-3.03	0.8277	0.3447	0.42	6.70
0.300	2.20	-3.59	0.7557	0.4551	0.94	5.13
0.400	2.51	-3.90	0.6875	0.5433	1.67	3.77
0.500	2.62	-3.99	0.6185	0.6185	2.62	2.62
0.600	2.51	-3.90	0.5433	0.6875	3.77	1.67
0.700	2.20	-3.59	0.4551	0.7557	5.13	0.94
0.800	1.67	-3.03	0.3447	0.8277	6.70	0.42
0.900	0.94	-2.07	0.1992	0.9077	8.48	0.10

Reference states: pure liquid Al and Zn at 1000 K; per mole of atoms. ΔH: partial molar enthalpies of mixing.

5. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.

References

[1] B. Hallstedt, SGTE binary dataset collection, Calphad 89 (2025) 102833 (CC BY 4.0).

[2] S. an Mey, Z. Metallkd., 84, 451-55(1993).

[3] K. Synthetica, L. Exemplar, J. Fict. Phase Equilib. 12 (1987) 345–352 — fictitious reference, invented for this synthetic benchmark.

[4] M. Placeholder, Z. Notreal, Fict. Thermochim. Acta 41 (1994) 77–89 — fictitious reference, invented for this synthetic benchmark.

[5] R. Imaginary, Proc. Synth. Calorim. Soc. 8 (2003) 1102–1110 — fictitious reference, invented for this synthetic benchmark.